

Course Project (Mathematics and Linear Algebra for AI)

Available: 10th March 2025

Due Date: 15th April 2025

Part 1 (MNIST Dataset):

Task 1: Consider the **Conv_AutoEncoder_Pytorch.ipynb** notebook. Make sure to familiarize yourself with the code. Run the code and check the different outputs.

Task 2: The embedding size is set to **2** in the notebook. Change the embedding size to **5, 10** and **20**. Produce the training loss using the **plot_training_loss()** function.

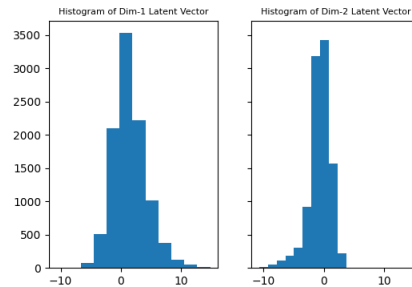
Task 3: Show samples of the generated images using the **plot_generated_images** function.

Task 4: Plot the latent space of the generated embeddings using the **plot_latent_space_with_labels** function for embedding sizes of **5, 10** and **20**.

Important Note 1: Make sure to use **PCA** to display the 2D projection of the generated embeddings.

Important Note 2: If you use **tSNE** projections in addition to PCA, you will get a bonus!

Task 5: Plot the histograms of the generated embeddings for embedding sizes of **5, 10** and **20** as we did for the case of embedding size **2** below:



Task 6: Use the AutoEncoder with embedding size **2** in inference. Feed each image in the test set and keep modifying the 1st dimension of the embedding from -10 to 10 in steps of 0.1 and display the reconstructed image as an animation (**gif file**).

Task 7: Use the AutoEncoder with embedding size **2** in inference. Feed each image in the test set and keep modifying the 2nd dimension of the embedding from -10 to 10 in steps of 0.1 and display the reconstructed image as an animation (**gif file**).

Part 2 (CelebA Dataset*):

*<https://pytorch.org/vision/main/generated/torchvision.datasets.CelebA.html>

Task 1: Modify the **Conv_AutoEncoder_Pytorch.ipynb** notebook to work with the **CelebA** dataset.

Task 2: Set the embedding size to **32** and **64**. Produce the training loss using the **plot_training_loss()** function.

Task 3: Show samples of the generated images using the **plot_generated_images**** function.

****Important Note:** You may need to modify the **plot_generated_images** function

Task 4: Plot the latent space of the generated embeddings using the **plot_latent_space_with_labels** function for embedding sizes of **32** and **64**.

Important Note 1: Make sure to use **PCA** to display the 2D projection of the generated embeddings.

Important Note 2: If you use **tSNE** projections in addition to PCA, you will get a bonus!

Task 5: Plot the histograms of the generated embeddings for embedding sizes of **32** and **64**.

Task 6: Use the AutoEncoder with embedding size **32** in inference. Feed each image in the test set and keep modifying the 1st dimension of the embedding from -10 to 10 in steps of 0.1 and display the reconstructed image as an animation (**gif file**).

Task 7: Use the AutoEncoder with embedding size **64** in inference. Feed each image in the test set and keep modifying the 2nd dimension of the embedding from -10 to 10 in steps of 0.1 and display the reconstructed image as an animation (**gif file**).